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B.Tech. course in Electronics and Telecommunication Engineering
Semester Examination, May - 2017

Subject: Embedded Systems Design
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Semester: VIII
Marks: 70

Instructions to students:

- 1) Figures to the right in square bracket indicates full marks.
- 2) PIC18F458 SFR list is provided at the end of question paper (Page 3).
- 3) Necessary data is given in the respective questions. If such data is not given it means its knowledge is a part of examination.
- 4) If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.
- 5) Q.1 is compulsory. Attempt any 5 questions from remaining.

Q. 1 Choose correct answer from given alternatives. [10]

- I. How many bits are required to store MOVLW 55H instruction?
a) 16 b) 08 c) 32 d) 24
- II. Which of the following value of TRISA register can be used to make RA6, RA5 & RA2 pins as output pins?
a) 00H b) 9CH c) 9DH d) 0CH
- III. Which register is used to access data indirectly from internal RAM of PIC18 microcontroller?
a) WREG b) FSR c) PIR d) LATA
- IV. In PIC18 microcontroller family, how many RAM locations are bit addressable?
a) 16 b) All c) 256 d) 128
- V. Maximum prescale value for Timer 0 is ____
a) 16 b) 128 c) 256 d) 32
- VI. When BRGH = 1, UART divides operating frequency by ____
a) 16 b) 32 c) 04 d) 08
- VII. In ARM state which registers are orthogonal?
a) r0 – cpsr b) r0 – r14 c) r0 – r15 d) r0 – r13
- VIII. Which of the following instruction is invalid?
a) MOV r0, #0xFF b) MOV r0, #0x00002200 c) MOV r0, #0x2300 d) MOV r0, #0x255
- IX. ARM Processor will be in Jazelle mode if ____
a) J=1, T=0 b) J=0, T=1 c) J=1, T=1 d) J=0, T=0
- X. In ARM instructions how many bits are used to represent conditional execution part of a instruction?
a) 04 b) 05 c) 02 d) 01

Q. 2 Attempt any two questions

- a) Write a program for PIC18 microcontroller to perform the following operations **[6]**
 - i) Keep monitoring the RB2 bit until it becomes HIGH;
 - ii) When RB2 becomes HIGH, write value 45H to Port C, and also send a HIGH-to-LOW pulse to RD3

- b) Assume that 10 8-bit numbers are stored from 40H to 49H internal RAM locations. Write a program for PIC18 microcontroller to add these numbers and store lower byte of 16 bit result at 50H and higher byte at 51H [6]
- c) Explain how ARM instruction set differs from the pure RISC definition? [6]

Q. 3 Attempt any two questions

- a) Write a program for PIC18 microcontroller to move values of block 50H - 55H to another block of memory locations 40H - 45H. Create and use a macro by any name that will transfer specified file registers data to WREG. [6]
- b) Assume that XTAL = 10 MHz for PIC18 microcontroller. Find the clock period fed into Timer0 if a prescaler option of 256 is chosen. Also find out what is the largest time delay we can get using this prescaler option for 16 bit mode of Timer 0? [6]
- c) Explain ARM7 3 - Stage pipelining with suitable example. [6]

Q. 4 Attempt following questions.

- a) Write a C program for PIC18 microcontroller to generate a square wave of 10Hz frequency on pin RB4. Use Timer0, 16 bit mode, the 1:4 prescaler to create the delay. Assume XTAL=10MHz. <https://www.batuonline.com> [6]
- b) Explain all Barrel Shifter operations with suitable examples of each. [6]

Q. 5 Attempt any two questions

- a) Assume that 8 switches and 8 LEDs are connected to PORTC & PORTD of PIC18 microcontroller respectively. Write a program to generate a square wave of 2 KHz on pin RB5, while at the same time transfer data from PORTC to PORTD. Assume that XTAL = 10 MHz. [6]
- b) Explain 32 bit encoding format of ARM data processing instructions in detail with neat diagram. [6]
- c) Explain different addressing modes for load-store multiple instructions of ARM with suitable example of each [6]

Q. 6 Attempt any two questions

- a) Explain following instructions with suitable example of each [6]
 - i) MVN ii) RSB iii) ORR
- b) Explain SWI instruction with 32 bit encoding format and a suitable example. [6]
- c) Explain following registers of LPC2148 : [6]
 - IOSEL0 ; IO1DIR; IO1SET;

- Q. 7** Interface 16x2 LCD to LPC2148 microcontroller. Write a C program to display a message "Dua me yaad rakhna..." on LCD continuously. [12]

Believe you can and you are halfway there...

All The Best

Pic18F458 SFR Data Sheet

File Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	
FDfh	INDF2	Uses contents of FSR2 to address data memory - value of FSR2 not changed (not a physical register)							
FDEh	POSTINC2	Uses contents of FSR2 to address data memory - value of FSR2 post-incremented (not a physical register)							
FDDh	POSTDEC2	Uses contents of FSR2 to address data memory - value of FSR2 post-decremented (not a physical register)							
FDCh	PREINC2	Uses contents of FSR2 to address data memory - value of FSR2 pre-incremented (not a physical register)							
FD8h	PLUSW2	Uses contents of FSR2 to address data memory - value of FSR2 pre-incremented (not a physical register) - value of FSR2 offset by W							
FDAh	FSR2H	—	—	—	—	Indirect Data Memory Address Pointer 2 High			
FD9h	FSR2L	Indirect Data Memory Address Pointer 2 Low Byte							
FD8h	STATUS	—	—	—	N	OV	Z	DC	C
FD7h	TMR0H	Timer0 Register High Byte							
FD6h	TMR0L	Timer0 Register Low Byte							
FD5h	T0CON	TMR0ON	T08BIT	T0CS	T0SE	PSA	T0PS2	T0PS1	T0PS0
FCFh	TMR1H	Timer1 Register High Byte							
FCeh	TMR1L	Timer1 Register Low Byte							
FCDh	T1CON	RD16	—	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON
FCCh	TMR2	Timer2 Register							
FCBh	PR2	Timer2 Period Register							
FCAh	T2CON	—	TOUTPS3	TOUTPS2	TOUTPS1	TOUTPS0	TMR2ON	T2CKPS1	T2CKPS0
FAFh	SPBRG	USART1 Baud Rate Generator							
FAEh	RCREG	USART1 Receive Register							
FADh	TXREG	USART1 Transmit Register							
FACh	TXSTA	CSRC	TX9	TXEN	SYNC	—	BRGH	TRMT	TX9D
FABh	RCSTA	SPEN	RC9	SREN	CREN	ADDEN	FERR	OERR	RC9D
FA5h	IPR3	IP0P	WAKP	ERRIP	TXB2P	TXB1P	TXB0P	RXB1P	RXB0P
FA4h	PIR3	IP0F	WAKF	ERRIF	TXB2F	TXB1F	TXB0F	RXB1F	RXB0F
FA3h	PIE3	IP0E	WAKE	ERRIE	TXB2IE	TXB1IE	TXB0IE	RXB1IE	RXB0IE
FA2h	IPR2	—	CMIP	—	EEIP	BCLIP	LVDIP	TMR3IP	ECCP1IP ⁽¹⁾
FA1h	PIR2	—	CMIF	—	EEIF	BCLIF	LVDIF	TMR3IF	ECCP1IF ⁽¹⁾
FA0h	PIE2	—	CMIE	—	EEIE	BCLIE	LVDIE	TMR3IE	ECCP1IE ⁽¹⁾
F9Fh	IPR1	PSPIP	ADIP	RCIP	TXIP	SSPIP	CCP1IP	TMR2IP	TMR1IP
F9Eh	PIR1	PSPIF	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF
F9Dh	PIE1	PSPIE	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE
FFBh	PCLATU	—	—	bit21 ⁽²⁾	Holding Register for PC<20:16>				
FFAh	PCLATH	Holding Register for PC<15:8>							
FF9h	PCL	PC Low Byte (PC<7:0>)							
FF8h	TBLPTRU	—	—	bit21 ⁽²⁾	Program Memory Table Pointer Upper Byte (TBLPTR<20:16>)				
FF7h	TBLPTRH	Program Memory Table Pointer High Byte (TBLPTR<15:8>)							
FF6h	TBLPTRL	Program Memory Table Pointer Low Byte (TBLPTR<7:0>)							
FF5h	TABLAT	Program Memory Table Latch							
FF4h	PRODH	Product Register High Byte							
FF3h	PRODL	Product Register Low Byte							
FF2h	INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF
FF1h	INTCON2	RBPUI	INTEDG0	INTEDG1	—	—	TMR0IP	—	RBIP
FF0h	INTCON3	—	INT1IP	—	—	INT1IE	—	—	INT1IF

*****END*****